

Pecking Order Theory of Capital Structure: Empirical Evidence for Listed SMEs in India

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Abstract

The purpose of our study is to empirically examine the relevance of pecking order theory (POT) in explaining the capital structure choices made by the listed small and medium enterprises (SMEs) in emerging capital markets. To do so, we use panel data regression on five years of data from 2015 to 2019 of 82 listed SMEs in India. In pursuit of robust results to test the theory, the study uses three econometric models: pooled ordinary least squares (pooled OLS), fixed effects (FE) regression and two-stage least squares (2SLS). Profitability, liquidity, growth, tangibility and non-debt tax shield are the independent variables, size is the controlled variable and financial leverage is the dependent variable. The pooled OLS and FE models provide biased estimates due to the presence of endogeneity. The 2SLS estimates overcome endogeneity in the explanatory variable non-debt tax shield by using tangibility as an instrument. The 2SLS provides a substantial improvement over pooled OLS and FE results. The results indicate that the explanatory variables, namely, profitability, liquidity, non-debt tax shield and size, support the POT. However, the growth and tangibility do not support the POT for listed SMEs. Overall, the results of our study are inclined towards the POT, suggesting that ease of access takes priority in financing decisions by SMEs. Careful consideration of country-specific factors will allow the results to be generalized to other emerging capital markets.

Key Words

Capital Structure, Pecking Order Theory, Endogeneity, Listed SMEs, India

Introduction

Small and medium enterprises (SMEs) play a significant role in the economic development of the nation. In Asia, SMEs and micro-enterprises constitute 96% of the total enterprises (Pinho et al., 2018). SMEs ensure the smooth functioning of economies, provide large-scale employment opportunities and lead nations towards growth. Therefore, SMEs are the key contributors to employment generation in India. However, one of the factors limiting the growth of SMEs is the inaccessibility of timely funds to finance their expansion plans. Consequently, the determination of capital structure procedures for SMEs becomes imperative. Capital structure is the outcome of the financing choices of the firm. Several studies focus on the capital structure choices made by firms leading to the development and testing of capital structure theories. In literature, most of the studies focus on large publicly listed companies in

developed economies (Allini et al., 2018). Studies on SMEs are mostly limited to privately held SMEs. Listing of SMEs on SME exchanges is a recent phenomenon in developing countries, including India. Listing of SMEs has resulted in notable status to these entities. However, literature on capital structure choices by publicly listed SMEs is limited. Over the decades, SMEs have attracted research interest on the determination of capital structure, which is derived from the fact that capital structure theories were not specifically designed for SMEs (Ang, 1991). The study by Michaelas et al. (1999) attempts to relate to the diverse theoretical characteristics of SMEs. Beck et al. (2005) conclude that smaller and new firms face greater financing problems and have less access to external finance, particularly the bank credit. Studies also specify firm's explicit characteristics describe dissimilarity in the capital structure choices of SMEs more reasonably than the country-specific

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facts (Daskalakis & Psillaki, 2008). Pecking order theory (POT) is correlated to the capital structure, which postulates that managers follow hierarchy to select sources of finance. This study aims to empirically examine the capital structure of publicly listed SMEs in India in the context of POT. In India, the SME sector is more concentrated towards the manufacturing sector than the service sector. Nearly 90% of the total industrial units in India fall under the SME manufacturing sector. Of the total exports, 45% of the industrial output and 40% of total output is from SMEs. Thus, the significant contribution by the SMEs to the economy has attracted the interest of policymakers and industrial bodies towards the development of SMEs. Motivated by the above issues, this study explores whether the POT can explain the capital structure choices of listed SMEs. This study is an attempt to empirically investigate the rationality of POT in capital structure determinants for listed manufacturing SMEs due to their noteworthy contributions to the Indian economy.

The inferences of the study are based on five years of data from 82 listed SMEs on the Bombay Stock Exchange (BSE) SME exchange of India. The pretext of capital structure as a research theme has its premise from the listing of SMEs. Listing of SMEs stimulated inflow of funds from capital as a supplementary source of financing, which otherwise was mainly reliant on owners' equity and bank finance. Listing has initiated reconsiderations of SMEs' capital structure mix and policies (Kulkarni & Chirputkar, 2014). The study affirms that POT claims persist due to the prevalence of information asymmetry between the SME owners and external providers of funds (Ang, 1991; Frank & Goyal, 2003; Holmes & Kent, 1991; Thampy, 2010). Information asymmetry increases the costs of capital for firms and persuades pecking order behaviour among firm managers. The manager's intent to eliminate information asymmetry and reduce costs motivates internal financing selection over external funds (Adair & Adaskou, 2015). The vested interest in wealth maximization and controlling power are the underlying central themes that direct the SME managers to implement POT. Debt is favoured as additional capital as it protects the dilution of ownership structure and offers a tax advantage. External equity is the last resort for SME financing. Although numerous financial theories have been developed to explicate the capital structure that a firm adopts, POT is more appropriate for identifying the capital structure than other theories for SMEs (Michaelas et al., 1998; Osei-Assibey et al., 2012). In India, a significant portion of the literature on capital structure focuses on large listed corporate enterprises, whereas limited studies exist for SMEs. This study aims to address the research gap in the finance literature addressing the research questions: (a) To what extent does POT subsists for listed SMEs in India, and (b) does POT rationalizes and assists in explaining the capital structure of listed SMEs in India.

This study attempts to contribute to the extant literature on capital structure from a developing country perspective,

addressing the arguments on financing behaviour of Indian listed SMEs centred around the POT. The study contributes to the literature in multiple ways. First, the study addresses the capital structure decisions made by publicly listed SMEs in developing economies, which is scarce in the literature. Listing of SMEs is a recent phenomenon in emerging markets, and thus the study contributes in terms of a different sample of Indian SMEs firms and current timeline. Second, the study supports the existing literature on POT for SMEs by testing three different regressions to discover a robust 2SLS model. Lastly, the study uncovers the impact of endogeneity of the variables used in the capital structure literature. This study intends to produce novel research findings within the constraints of a limited sample size of listed SMEs. The sample is limited because only 220 SMEs are listed on the BSE SME exchange in January 2021, limiting this study's overall population. Finally, the concepts and conclusions of this study will open up new avenues of discussion and research on the impact of listing on the capital structure of SMEs.

The remainder of this study is organized to discuss the capital structure theories, literature review and specification hypotheses, methodology, econometric analysis and the conclusion.

Capital Structure Theories and Literature Review

This section summarizes the capital structure theories and the previous literature leading to discovering the research gap, extracting constructs and finally defining the study's variables.

The proposition of capital structure theories was propounded by Modigliani and Miller (1958) on the irrelevance of capital structure on firm value. The original Modigliani and Miller's Theorem (1958) proposes an efficient market structure, no transaction costs and bankruptcy costs with free flow of information in the market. In 1963, Modigliani and Miller built in the effect of taxes into their model to mark it as realistic. Further, combining securities in capital structure would increase the firm's value was an undefined assumption that the Modigliani and Miller's theorem emphasized. Empirical evidence affirms that capital structure is critical as it significantly impacts the firm (Myers & Majluf, 1984). The existence of a universal theory can be argued in the context of numerous factors and circumstances that would explain a firm's behaviour (Myer, 2001). A widely cited theory of capital structure decisions is the POT, which states that firms' financing behaviour is based on a hierarchy of choices of financial sources (Myers, 1984). Myers and Majluf (1984) proclaim that firm managers prefer internal funds first, external debt, and merely in exceptional settings issue new equity. On the other hand, the trade-off theory emphasizes an optimal capital structure that firms can attain by harmonizing benefits and the costs of debt (Myers, 1984).

The literature on capital structure discusses various theories: POT, trade-off theory, signaling theory, agency theory and the market timing theory (Chaklader & Chawla, 2016). POT is widely considered in academic research (Buvanendra et al., 2016). Proposed by Donaldson (1961), adapted and advocated by Myers and Majluf (1984), the POT advocates the rubrics of adverse selection problems and information asymmetry. Information asymmetry leads to overstated financing costs for the firms to compensate for adverse selection by the investors (Rao et al., 2019). The POT signifies a hierarchy in choice of funds based on quick availability and postulates the internal financing as first, then debt till its depletion and lastly, the use of equity financing (Myers, 1984; Myers & Majluf, 1984; Shyam-Sunder & Myers, 1999). For instance, an empirical investigation of the theoretical model of POT by Shyam-Sunder and Myers (1999) produced evidence supporting the claim that firms rely on internal sources of funds. Financial deficit persuades firms to consider external financing sources in a preferential manner considering secured debt, consequently by risky debt and finally external equity.

The core of capital structure underlies the capital mix choices, the accessibility of different sources of funds and the firm's behaviour in making capital structure choices. Traditionally, SMEs have over-relied on owners' funds, informal sources and bank finance for their financial needs (Kulkarni & Chirputkar, 2014). The owners' fund is confined to the owners' wealth, loans from relatives and friends, depending on the promoter's reputation in the market. Access to bank finance is limited on the availability of collateral and also to unsecured loans. The inherent risks and information asymmetry make loans expensive for SMEs than their larger counterparts (Kulkarni & Chirputkar, 2014). The long-standing financial hurdles of SMEs for growth and expansion are applicable in the present scenario in the developing nations (Boocock & Wahab, 2001; Kulkarni & Chirputkar, 2014; Thampy, 2010). Nevertheless, there are remarkable efforts by policymakers to decrease the financial accessibility gap for SMEs. Reasons attributable to SMEs' financial hurdles are lending bias, uneven financing framework, underdeveloped capital markets (Baker et al., 2020) and perceived incidence of information asymmetry (Frank & Goyal, 2003; Thampy, 2010). Information asymmetry drives away potential SME investors and increases the cost of finance due to the premium charged by investors for their risky investment selection. From the firms' perspective, the prevalence of information asymmetry, where owners and managers have implicit information than external investors, leads to pecking order behaviour by the firm managers. The study addresses the financing decisions of SMEs, recognizing the existence of awful asymmetries of information amid agents. Literature evidence studies in developed countries on large corporate enterprises focusing on the factors influencing a firm's capital structure (Bancel & Mittoo 2004; Bevan & Danbolt, 2002; Shyam-Sunder & Myers, 1999; Titman & Wessels, 1988). Studies that examined capital structure anomalies

emphasized country-specific factors such as institutional framework and culture. These factors were dissimilar in nature for emerging markets than developed economies (Belkhir et al., 2016; Antonczyk & Salzmann, 2014). Moreover, studies conducted for SMEs in emerging markets are limited and focused on unlisted privately held firms (Baker et al., 2020; Mac an Bhaird & Lucey, 2014; Ramcharran, 2017; Rao et al., 2019). The extant literature reiterates the arguments on the presence of POT for SMEs (Daskalakis & Psillaki, 2008; Degryse et al., 2012; Mateev et al., 2013; Rao et al., 2019).

The practice of SME listing is a new phenomenon in developing countries. In India, the listing of SMEs started with the launch of the SME-specific EMERGE platform by the National Stock Exchange and the BSE SME platform by the BSE in the year 2012. Listing of SMEs offers advantages of equity capital infusion, access to other avenues of raising capital, convenient exit routes to private equity investors, higher visibility and good governance. Since the capital structure is an alignment of debt and equity, the listing has positively impacted the leverage levels of SMEs. However, listed enterprises in the efficient and developed capital markets may witness less evidence on the traditional hierarchy of the POT due to the dissemination of information arising out of growth opportunities of the enterprises and discretionary capital choices of agents which are not aligned with the shareholders' interest. Managers may show aversion to relinquishing their rights of control in the business. Previous studies indicate that POT may hold for the rights issues where current shareholders are not wealthy enough to purchase additional stock to maintain their existing ownership control (Cressy & Olofsson, 1997; Hamilton & Fox, 1998; Jordan et al., 1998). The study by Kanojia et al. (2020) on the pattern of financing of 428 listed SMEs in India indicates POT. Further, the study discusses that smaller SMEs were dependent on external financing to increase the company's size, which contradicts the studies from developed economies. The study confirmed the existence of information asymmetry in the Indian SMEs confining to capital constraints. Capital structure decisions are subject to various metrics. The firm's growth is measured by the current year's revenue change from the previous year's revenue. Firms subject to fewer growth estimates depend on debt (Jensen & Meckling, 1976; Stulz, 1990). Liquidity indicates the fulfilment of short-term obligations; the trade-off assumption estimates a positive connection between liquidity and debt (Degryse et al., 2009). The magnitude of liquidity reflects the risk of bankruptcy affecting the capital structure (Baxter, 1967; Warner, 1977). Asset structure defines the tangibility and existence of collaterals. Collateral moderates the information asymmetry for the firms signifying a positive impact of POT (Degryse et al., 2009; Titman & Wessels, 1988).

The pecking order principle has been preferred as a traditional subject in previous studies. However, a few studies claim that the theory of matured pecking order does not

hold for transitional economies. Overall, the research offers an outlook of Indian listed SMEs' financial trends and thus provides conclusive proof of the prevalence of POT for smaller firms in developing countries (Singh, 1995). Lemmon and Zender (2010) discovered that POT explained the observed financing behaviour of a firm to a great extent. A study of Brazilian SMEs evidenced a negative influence of leverage on profitability, which further supports the POT argument that firms prefer internal financing to external financing (Forte et al., 2013). Further, Degryse et al. (2012) affirmed that Dutch SMEs favoured debt finance as a subsequent source of finance after draining their internal funds, thereby validating the arguments of POT. The high cost of debt for SMEs (Kulkarni & Chirputkar, 2014) compared to larger counterparts forced SMEs to plow their profits to reduce and eliminate the high-interest burden of debt (Degryse et al., 2012). SMEs are subject to financial constraints and rely exceptionally on internal funds equated to other sources of finance, enabling the theoretical framework of POT to be vital for SMEs (Mateev et al., 2013). An empirical study by Choudhry and Sundaram (2013) on high growth Indian firms from 1991 to 2009 suggests that the implications of POT as per Myers (1984) do not quite hold in the Indian scenario. The study by Chaklader and Chawla (2016) on factors determining the capital structure of Indian firms during global slowdown 2008–2015 emphasizes the trade-off theory and concludes liquidity as the only independent variable which supports the POT. The study by Chauhan (2016) affirms that listed firms' leverage supports the prediction of the pecking order, and firms do not realign their debt ratios to withstand the drift caused by stock movements in debt ratios. The study by Rao et al. (2019) on SMEs using the two-step estimation method evidenced that capital structure decisions of SMEs align with the conventions of POT. This result, however, differed for larger firms.

Frank and Goyal (2003) discovered that POT does not hold true for firms with significant information asymmetry. Okpara and Yartey (2011) stressed the prominence of external debt more than the internal funds for SMEs in Ghana, which contradicts the proclamations of POT. The controlling behaviour of SME owners is one of the potential reasons to choose debt over other sources of funds (Benkraiem & Gurau, 2013). While studying firms from India, South Africa and China, Kannadhasan et al. (2018) discovered that POT is unsuitable for discussing capital structure. Similarly, for Indian firms, POT was not found to be much relevant in capital structure decisions (Ghose & Kabra, 2018). The literature thus provides some contradictory evidence in the context of capital structure decisions made by SMEs. Kannadhasan et al. (2018) tested the capital structure theories on a stand-alone basis and an integrated framework using advanced dynamic panel data of 1,183 firms from 1999 to 2016. The study affirms that trade-off theory explains the capital structure decisions better than the POT.

The literature on POT reflects mixed outcomes in a country-specific context. Very few empirical studies

evidence that trade-off theory outperforms POT in developed countries. Studies also indicate that POT assumes no targeted optimal ratio. Contrarily, the debt ratio is an aggregate outcome of hierarchical financing choices over a while. Extant literature also specifies that a firm's financing decisions are influenced by its characteristics. Besides, the literature signifies that the empirical evidence on capital structure decisions of listed SMEs in developing economies is scarce. Various studies in the extant literature have widely discussed that SMEs' information asymmetry and lack of transparency lead to pecking order behaviour. However, the validity of these claims is untested for listed SMEs, particularly in the emerging markets context. Literature on POT in capital structure decisions is prevalent for developed countries and mainly for large corporate enterprises. Therefore, it is crucial to test the applicability of the POT for listed SMEs in developing economies, which is a considerable gap as of today. Therefore, the study attempts to test the POT on the listed SMEs in India, which is considered a developing economy. Besides, previous studies in this context leave us with a diverse set of evidence. This study examines the relevance of POT in capital structure decisions on Indian listed SMEs and provides an appropriate model when considering financing decisions. In this way, our study addresses the identified research gaps.

Specification of Hypotheses

Literature provides key independent variables for determination of capital structure choices: profitability (Chaklader & Chawla, 2016; Forte et al., 2013; Jauhari & Madan, 2007; Myers & Majluf, 1984; Rao et al., 2019), liquidity (Chaklader & Chawla, 2016; Chakraborty, 2010; Rao et al., 2019), growth (Chakraborty, 2010; Rao et al., 2019) and non-debt tax shield (Chaklader & Chawla, 2016; Chakraborty, 2010; Kaur & Rao, 2009; Rao et al., 2019), tangibility (Chaklader & Chawla, 2016; Rao et al., 2019; Shanmugasundaram, 2008) and size (Chaklader & Chawla, 2016; Chakraborty, 2010; Daskalakis & Psillaki, 2008; Rao et al., 2019).

Leverage (LEV)

Leverage is indicated by the debt-to-equity ratio and explained variables in the study. The choice of the variable leverage to test the POT is based on a literature review and is referenced by multiple previous studies. One of the notable articles includes the studies from Titman and Wessels (1988) on examining the theories and factors of capital structure. In addition, recent studies from Chaklader and Chawla (2016), Rao et al. (2019), Baker et al. (2020) and Kanojia et al. (2020) examine the capital structure choices in the Indian context.

Profitability (ROE)

Firms with stable profits tend to avoid debt financing. In the case of SMEs, profit may be used to pay off debts to

evade the burden of high interest rates. This is as per the financial hierarchy signified by POT. Profitability (*ROE*) is considered an independent variable, which demonstrates an inverse relation with the leverage, and as per the postulates of POT (Chaklader & Chawla, 2016; Forte et al., 2013; Myers & Majluf, 1984; Rao et al., 2019). Based on the inferences from the studies mentioned above, and to check their applicability for listed SMEs, we propose the hypothesis, H_1 as follows:

H_1 : Profitability (*ROE*) is inversely related to leverage (*LEV*).

Liquidity (LIQ)

As per the postulates of POT, firms with high liquidity attempt to avoid debt and make use of internal funds. The argument that the cost of debt is high for listed SMEs (due to inherent risk and information asymmetry), liquidity should exhibit an inverse relationship with leverage. Therefore, liquidity was considered an explanatory variable to test the POT with favourable results (Chaklader & Chawla, 2016; Chakraborty, 2010; Rao et al., 2019). Therefore, we propose the hypothesis, H_2 as follows:

H_2 : Liquidity (*LIQ*) is inversely related to leverage (*LEV*).

Growth (RGS)

The growth of any business requires capital. Therefore, the issue of equity to fund growth opportunities may signal a lack of confidence in the business to the outside investors, while debt capital demonstrates confidence in the business prospects. Hence, as per the arguments of POT, a growing firm would avoid the issue of equity and fund the business from debt capital. These results are similar to the studies conducted by Chakraborty (2010) and Rao et al. (2019). Therefore, hypothesis H_3 is framed as follows:

H_3 : Growth (*RGS*) is positively related to leverage (*LEV*).

Non-Debt Tax Shield (NDTS)

DeAngelo and Masulis (1980) proposed a model to incorporate the effect of non-debt-related corporate tax shields into the capital structure. According to their proposal, tax deductions attributable to depreciation and tax credits on investments may substitute the tax benefits arising from debt financing. Consequently, firms with sizeable non-debt tax shields prefer less debt. Accordingly, Rajagopal (2011) found a negative association between *NDTS* and capital structure, favouring the POT. However, literature also reveals contradictory results on this relationship (Kaur & Rao, 2009; Mallikarjunappa & Goveas, 2007). Therefore, we propose the hypothesis, H_4 as follows:

H_4 : Non-debt tax shield (*NDTS*) is inversely related to leverage (*LEV*).

Tangibility (TAN)

Firms with sizeable tangible assets can generate significant collateral to borrow debt capital to fulfil their financial needs. However, according to POT, a higher level of tangible assets reduces asymmetric information, allows the issue of shares at fair to high prices, ultimately enabling the firm managers to avoid debt funds (Harris & Raviv, 1991). Furthermore, consistent with POT, several studies deduce a negative relationship between tangibility and leverage (Butt et al., 2013; Rakshit & Sarkar, 2013). Therefore, we propose the hypothesis, H_5 as follows:

H_5 : Tangibility (*TAN*) is negatively related to leverage (*LEV*).

Size (SIZ)

Size is a crucial variable in studies of factors impacting the capital structure (Abor & Biekpe, 2009). Frank and Goyal (2003) proposed that size of the firm may impact the borrowing capacity. It may be easier for larger firms to acquire debt capital than their smaller counterparts. Hence, the variable size is controlled in this study and defined as a natural log of total assets. However, larger firms may have stable earnings, and hence, they tend to avoid debt as a choice of finance (Rao et al., 2019). We propose the hypothesis, H_6 as follows:

H_6 : Size (*SIZ*) is negatively related to leverage (*LEV*).

Research Methodology

The research aims to test and reaffirm the results of capital structure theories in the context of POT (Collis & Hussey, 2003). The type of research proposed for the study is quantitative. Quantitative research involves data aiming to test a hypothesis and generalize results from a sample to a larger known population.

Sample and Methods

The population of the study consists of 220 SMEs listed on BSE SME exchange for the year 2020. Approximately 47% of the SMEs listed on the BSE SME exchange are from the manufacturing sector, which constituted the sample set for the study. The services and financial sectors SMEs were excluded from the study because of the differences in their capital structure. The debt-to-equity ratio varies between industries due to capital intensity levels and the nature of business. Studies state that the average debt-equity ratio of small companies is more than the average debt-equity ratio of large companies in India. Dissimilarity in capital structure in different industries are subject to investment decisions. A cluster sampling approach was adopted for the selection of 82 listed SMEs from the manufacturing sector. Secondary financial data for five years (2015–2019) are extracted from the financial reports for the selected firms. The panel data for the study consist of

82 cross-sections of 5 years of data, resulting in 410 observations. Since the study comprised multiple firms for analysis for five years, panel data have been used.

The purpose of the study is to examine the significance of SME-specific indicators like leverage, liquidity, growth in sales, profitability and non-debt tax shield in determining the capital structure of companies in the manufacturing sector listed on the BSE SME stock exchange. This study uses panel data analysis to consider the transversal information and the time period to determine the effect of firm-specific factors on capital structure. Panel data methodology has the advantage of considering the individual characteristics of each SME. Panel data blends the inter-individual differences and intra-individual dynamics to generate more accurate inferences of model parameters (Hsiao, 2005). Furthermore, regression using panel data can deal with the problem of omitted variable bias, even when there is a lack of information on variables that correlate with both the regressors and the dependent variable. The iteration process included pooled ordinary least squares (OLS), fixed effects (FE) regression and the two-stage least squares (2SLS) method to examine the POT empirically.

Variables

Table 1 lists the variables of this study and elucidates their measurements.

Econometric Analysis and Findings

This section describes the data, tests for Gauss–Markov assumptions, specifies the models and presents the econometric results.

Descriptive Statistics

Table 2 presents the descriptive statistics of the sample. The descriptive statistics indicate within the firm and intra-firm variation. The dependent variable *LEV* indicates that few listed SMEs are debt-free, while the maximum value of 4.73 indicates the presence of high leverage. The mean value of *ROE* does not indicate good returns to the shareholders. *ROE* is affected due to the rise in the interest cost declining the profits for highly levered. The descriptive statistics indicate low to high liquidity for the select firms on the removal of outliers. A large spread in values of *RGS* is found for the select SMEs. The variable *NDTS* ranges from 0.0007 to 0.2509. The variable tangibility also depicts variation between the firms within the same industry. Firm size is the control variable in the model. No significant outliers impacted the variables except for liquidity and growth. The removal of outliers was undertaken based on the normal score calculated from the mean (μ) and standard deviation (σ). The upper limit was set to $(\mu + 3*\sigma)$ and the lower limit to $(\mu - 3*\sigma)$.

Correlation Matrix

Correlation analysis was undertaken to measure the correlation between the variables and infer the association between the sets of variables. Table 3 indicates the correlation matrix of the independent variables used in the empirical model. The correlation coefficients between the independent variables are not high, except for the value 0.62 between *TAN* and *NDTS* due to the presence of total assets as a common factor while measuring the variables *TAN* and

Table 1. List of Variables and Their Measurement.

Types	Variables	Measurements
Dependent variable	Leverage (<i>LEV</i>)	$\frac{\text{Current liabilities} + \text{Long term debts}}{\text{Total assets}}$
Independent variables	Liquidity (<i>LIQ</i>)	$\frac{\text{Current assets}}{\text{Current liabilities}}$
	Non-debt tax shield (<i>NDTS</i>)	$\frac{\text{Depreciation} + \text{Amortization}}{\text{Total assets}}$
	Rate of growth in sales (<i>RGS</i>)	$\frac{(\text{Current year sales} - \text{Previous year sales}) \times 100}{\text{Previous year sales}}$
	Tangibility (<i>TAN</i>)	$\frac{\text{Fixed assets}}{\text{Total assets}}$
	Profitability (<i>ROE</i>)	$\frac{\text{Profit after tax}}{\text{Total equity}}$
Control variable	Size (<i>SIZ</i>)	$\ln (\text{Total assets})$

Source: The authors.

Table 2. Descriptive Statistics.

Variables		Mean	Std. Dev.	Min.	Max.	Observations
LEV	Overall	1.45	0.49	0.31	4.74	<i>N</i> = 410
	Between		0.40	0.51	2.80	<i>n</i> = 82
	Within		0.29	0.18	4.35	<i>t</i> = 5
ROE	Overall	0.07	0.37	−6.94	0.96	<i>N</i> = 410
	Between		0.16	−1.12	0.37	<i>n</i> = 82
	Within		0.34	−5.75	2.16	<i>t</i> = 5
LIQ	Overall	8.43	17.34	0.41	121.14	<i>N</i> = 410
	Between		12.68	1.23	61.52	<i>n</i> = 82
	Within		11.90	−41.37	90.01	<i>t</i> = 5
RGS	Overall	35.99	82.36	−160.00	411.72	<i>N</i> = 410
	Between		57.03	−152.60	293.10	<i>n</i> = 82
	Within		59.68	−301.31	365.36	<i>t</i> = 5
NDTS	Overall	0.04	0.03	0.00	0.25	<i>N</i> = 410
	Between		0.03	0.00	0.15	<i>n</i> = 82
	Within		0.01	−0.04	0.14	<i>t</i> = 5
TAN	Overall	0.30	0.23	0.00	0.92	<i>N</i> = 410
	Between		0.21	0.01	0.85	<i>n</i> = 82
	Within		0.10	−0.27	0.68	<i>t</i> = 5
SIZ	Overall	19.14	1.27	11.51	22.28	<i>N</i> = 410
	Between		1.09	16.32	21.36	<i>n</i> = 82
	Within		0.65	11.86	21.60	<i>t</i> = 5

Source: Results of statistical analysis conducted by authors through STATA.

Notes: *n* = sample size; *t* = time periods.

Table 3. Correlation Matrix of Independent Variables.

	ROE	LIQ	RGS	NDTS	TAN	SIZ
ROE	1					
LIQ	−0.0162	1				
RGS	0.0188	0.1455	1			
NDTS	−0.1133	−0.2648	−0.0212	1		
TAN	−0.0499	−0.3246	−0.0196	0.6245	1	
SIZ	−0.0073	−0.0275	−0.2159	0.0024	0.0985	1

Source: Results of statistical analysis conducted by authors through STATA.

NDTS. However, the variance inflation factors (VIFs) for independent variables were less than three. Therefore, all variables were considered for analysis in the study.

Regression Models

Pooled OLS, FE regression and 2SLS were used to examine the POT empirically. The variation in methods permits in identifying a robust model. However, due to the complex relationships of the variables used in the model, Gauss–Markov assumptions for OLS cannot be assumed and require checks. Besides, testing for firm FE in the case of panel data is an important consideration to select a suitable model between the FE and random effects.

Pooled OLS

The Gauss–Markov assumptions need to be satisfied for OLS estimation. Important assumptions to check are

random sample selection, linearity, no multicollinearity, homoscedasticity, normal distribution and homogeneity. The sample for the study was selected based on the cluster sampling technique and satisfied the first condition of random sample selection. The second assumption of linearity between the explanatory and explained variables is satisfied, excluding the firm's variable size. Hence, size is controlled by using the natural log of total assets to satisfy the condition of linearity. The correlation matrix of regressors shows no perfect correlation between them. In fact, the largest correlation coefficient value between *NDTS* and *TAN* is 0.62, an acceptable value. Plus, the VIFs for all independent variables, when tested separately, were found to be very low ($VIF < 3$). Further, the estimates using 2SLS, which employ instrumental variables, can deal with multicollinearity, if any. Besides, scatter plots were used to test the presence of heteroscedasticity. While

heteroscedasticity causes biased OLS estimates, it cannot substantially impact the 2SLS estimates. Next, given the sample size of 82 firms for five years each, the central limit theorem allows us to assume normality. Lastly, the presence of endogeneity may cause severe distortion of estimates, if unaddressed. Therefore, the 2SLS method was used, followed by the Durbin–Wu–Hausman test for endogeneity. Lastly, an instrument validity test was conducted.

Model 1

The pooled OLS model for the study:

$$LEV = \alpha + \beta_1 (ROE) + \beta_2 (LIQ) + \beta_3 (RGS) + \beta_4 (NDTS) + \beta_5 (TAN) + \beta_6 (SIZ) + \epsilon$$

Although all the assumptions of OLS cannot be realized in practice, a pooled regression was undertaken of the model, assuming homogeneity. However, using a pooled OLS regression, the unobserved discrete properties of listed SMEs are not measured; therefore, the resultant heterogeneity can impact the dimensions of the estimated parameters. However, it is possible to control variables of capital structure comparisons based on empirical evidence of the firms by using panel models of FE. Table 4 indicates the output results that were not found significant and had a low model R^2 (0.12).

Fixed Effects Model or Random Effects Model

In panel data analysis, the Hausman test, also known as the test for model misspecification, enables selecting between FE or a random effects model. The hypothesis for the Hausman test states that the FE model is the true model. The results indicated that the FE model is appropriate for the study, as evident from the output (p -value = 0.0003***,

where *** represents 1% level of significance). However, it was discovered that the overall R^2 -value was very low (0.03), reflecting omitted variable bias.

Model 2

The FE regression model,

$$LEV_{it} = \beta_1(ROE)_{it} + \beta_2(LIQ)_{it} + \beta_3(RGS)_{it} + \beta_4(NDTS)_{it} + \beta_1(TAN)_{it} + \beta_1(SIZ)_{it} + z_i + u_{it}$$

where $i = 1-82$ for the SMEs, $t = 2015-2019$, z_i captures the firm FE, and u_{it} is the error term.

Dealing with Endogeneity

The OLS estimator β is unbiased and consistent if the explanatory variables are exogenous [$E(\epsilon|X) = 0$]. This means that the error terms in the regression model are uncorrelated to the explanatory variables. The independent variables that are not exogenous are called endogenous variables. Endogeneity occurs due to three common reasons: measurement errors in variables, reverse causality and omitted variables (Wooldridge, 2010). Reverse causality occurs when both independent and dependent variables cause each other simultaneously. The omitted variables bias occurs because of correlated omitted variables and violates the assumption of zero conditional means. Thus, endogeneity is a severe problem and leads to biased OLS estimates, leading to faulty conclusions.

Dealing with endogeneity is an art as well as science. All measurement errors in variables should be avoided, there should be no reverse causality and no variables should be omitted from the model. This is unlikely in reality, and hence alternate methods to deal with endogeneity should be used. Two standard methods used to tackle

Table 4. Econometric Results of Pooled OLS, FE and 2SLS.

Variables		Model 1 (Pooled OLS)		Model 2 (FE)		Model 3 (2SLS)	
		Beta	P-value	Beta	P-value	Beta	P-value
Independent variables	ROE	-0.0612	.3260	0.0187	.6730	-0.1226*	.074
	LIQ	-0.0074***	.0000	-0.0042***	.0010	-0.0085***	.000
	RGS	-0.0002	.4770	0.0006**	.0210	-0.0002	.442
	NDTS	1.5703*	.0860	3.0692**	.0140	-6.7033***	.000
	TAN	-0.7158***	.0000	-0.1796	.3350	0.0865***	.000
	SIZ	-0.0464**	.0130	-0.1144	.0000	-0.0596**	.003
Tests	Wald test					46.96***	.000
	Durbin (score) χ^2					27.30***	.000
	Wu–Hausman F stat					28.75***	.000
R^2		$R^2 = 0.12$		R^2 Within = 0.15		$R^2 = 0.40$	
		Adjusted $R^2 = 0.11$		R^2 Between = 0.03		Adjusted $R^2 = 0.39$	
				R^2 Overall = 0.07			

Source: Elaborated by authors from the output of STATA.

Note: For 2SLS, Instrumented = NDTS, Instruments = ROE, LIQ, RGS, SIZ, TAN. *** $p < .01$, ** $p < .05$, * $p < .1$.

The theoretical expectations and the observed results based on 2SLS regression are presented in Table 5.

endogeneity in panel data are 2SLS and generalized methods of moments (GMM). GMM is more suitable and gives consistent results for sufficiently large data, and hence not suitable for our study. The 2SLS estimation method uses instrumental variables to correct endogeneity. This method involves using an instrumental variable that should be exogenous, correlated with the endogenous variable, has no independent effect on the explained variable and must be uncorrelated with the model error (Wooldridge, 2015). Endogeneity is model-specific and not variable. Potential endogenous variables are selected based on theoretical considerations and tested separately for endogeneity (Salkind, 2010).

Model 3

The 2SLS estimation with instruments consists of two steps. First, the endogenous regressor is regressed on the selected instrumental variables, and the regression residual is saved. Then, the explained variable is regressed on the residual from the first step instead of the endogenous regressor (Wooldridge, 2015). The two steps of the model for our study are represented as follows:

Step 1:

$$NDTS = \alpha + \beta_1 (ROE) + \beta_2 (LIQ) + \beta_3 (RGS) + \beta_4 (TAN) + \beta_5 (SIZ) + \epsilon$$

$$NDTS_{residual} = NDTS - NDTS_{predicted}$$

Step 2:

$$LEV = \alpha + \beta_1 (NDTS_{residual})$$

The Durbin–Wu–Hausman test is used to check the endogeneity of variables. We use the 2SLS method to deal with endogeneity. Based on the literature review, *LIQ* and *RGS* were selected to be purely exogenous. We tested *NDTS*, *TAN*, *ROE* and *SIZ* for endogeneity and found that the only variable that tests endogenous by applying the Durbin–Wu–Hausman test (at a 5% level of significance) is the variable *NDTS* when instrumented by *TAN*. *TAN* was selected as an instrument for two reasons. First, it is

correlated with *NDTS* by a value of 0.62 and; second, it was not independently able to predict the dependent variable *LEV*. Further, the roles of *NDTS* and *TAN* were reversed, and the endogeneity of *TAN* was not established. Further, the instrument *TAN* was found to be strong and valid, given by *F*-statistics of 115.94. The *R*² value of 0.40 and the Adj-*R*² of 0.39 indicate a significant improvement by using 2SLS. The results are plotted in Table 4.

Results of Econometric Analysis

The summary of the results of the study is presented in Table 5. The econometric analysis indicates that the pooled OLS and FE model produce biased results. The 2SLS, on the other hand, deals with the problem of endogeneity and provides an improved explanatory power. The results of pooled OLS, FE and 2SLS are presented in Table 5. However, the following discussion will use the estimates obtained from the 2SLS, which were the most robust of the three models.

Conclusion

In this study, an attempt is made to investigate the determinants of the capital structure of 82 listed SME firms on the BSE SME stock exchange. The investigation was conducted using panel data techniques, namely, FE. The empirical analysis indicates that liquidity, *RGS*, *ROE* and size are negatively associated with leverage. The output of 2SLS regression shows that the estimated coefficient of *ROE* is negative and insignificant (*p*-value > .05). The negative sign supports POT, and the results are as per the observations of Rao et al. (2019). In listed SMEs, there is an inverse relationship between profitability and leverage; it enhances the likelihood of financial hierarchy in the capital structure decisions. The effect of liquidity on leverage is negative and highly significant (*p*-value < .05). Therefore, the second hypothesis is accepted. The finding that liquidity variables support the POT indicates that listed SMEs with high liquid assets prefer internal financing to external funds. Interestingly, the situation indicates the optimal use of funds by listed SMEs. The result is inconsistent with the study conducted by Rao et al.

Table 5. Theoretical Expectations and Observed Results.

Variables	Relationship with <i>LEV</i>		Results
	Predicted	Observed	
<i>ROE</i>	–	–	The variable <i>ROE</i> follows POT.
<i>LIQ</i>	–	–	The variable <i>LIQ</i> follows POT.
<i>RGS</i>	+	–	The variable <i>RGS</i> does not follow POT.
<i>NDTS</i>	–	–	The variable <i>NDTS</i> follows POT.
<i>TAN</i>	–	+	The variable <i>TAN</i> does not follow POT.
<i>SIZ</i>	–	–	The variable <i>SIZ</i> follows POT.

Source: The authors.

(2019). The observations of *ROE* and *LIQ* generate exciting questions, the first being the claim of high-interest rates of debt by Kulkarni and Chirputkar (2014) for SMEs. Listed SME firms are subject to a certain level of information asymmetry and a high interest rate on formal credit. The coefficient of *RGS* is negative but insignificant (p -value $>.05$), which means that listed SMEs have taken the path of equity to fund the growth opportunities. This leads to the rejection of the third hypothesis. A risk-averse behaviour is found in listed SMEs despite prospects for growth firms are reluctant to fund through debt financing. The implication of *RGS* is different from the inference of Rao et al. (2019), indicating a difference in the behaviour of listed versus unlisted SMEs. The coefficient of the non-debt tax shield is negative and highly significant (p -value $<.05$), which claims to accept the fourth hypothesis. The results contradict the studies of Chaklader and Chawla (2016) and Rao et al. (2019). The tangibility coefficient is positive and highly significant (p -value $<.05$), rejecting the fifth hypothesis and contradicting the POT. This is consistent with earlier studies that tangible assets help build sound collateral to borrow debt and enable the managers to avoid issuing external equity. The firm's size is negative and significant (p -value $<.05$), rejecting the sixth hypothesis. Listed SMEs go for public issues and not for debt as the size of the firm increases. The result is as per the study of Rao et al. (2019). The inverse relationship between the size of listed SMEs and their capital structure indicates stable earnings, making debt financing a less rewarding choice. The variable size supports the POT. Overall, the study indicates that the listed SMEs are inclined towards the POT in emerging capital markets, which is in line with the results obtained by Kanojia et al. (2020). Finally, the R^2 value of 0.40 indicates an acceptable explanatory power of the 2SLS model; this study aims to examine the POT for listed SMEs in India is well achieved. There are a few important reasons and implications of the pecking order behaviour in listed SMEs. Though the listing has substantially transformed the financial choices for SMEs, information asymmetry has its consequences.

With access to capital market funds, SMEs were expected to no longer over-rely on banks for their financial needs and that the fixed liability of interest burden could be eliminated. SME owners are not ready to part with their ownership and issue only marginal equity as per listing norm requirements. Promoters' holding signifies the listing of a fraction of shares above the Securities and Exchange Board of India regulatory requirement. Increased compliance and information disclosures due to listing requirements permitted listed SMEs to access debt. Capital structure choices of listed SMEs do not differ significantly from the earlier studies, which primarily support the pecking order behaviour. The literature specifies that firm-specific factors decide the leverage ratios, while non-firm-specific factors relatively have a limited impact. The study has important policy implications for financing decisions.

These determinants can be considered as a benchmark for financial decisions related to capital structure for SMEs. Besides, the study will provide critical insights to the practitioners and academics towards fine-tuning the financing choices of listed SMEs.

Like all other studies, our study has some limitations. First, to analyse the capital structure choices of listed SMEs, the study considered only one theory, the POT. However, the purpose of considering only theory is to ensure clarity in observations and avoid confusion of discourse and arguments due to divergent assumptions of multiple theories. The other limitation is that the research originates from young SME exchanges in India, which are not fully developed. However, the study unlocks numerous avenues of future research on capital structure for listed SMEs. Further studies can be conducted based on other capital structure theories like trade-off theory, market timing theory and so on. Also, studies on the capital structure for firms from the services sector can be considered for empirical investigation. Diversification of risk, increased options for growth after listing, improved credit rating and enhanced financial valuation and sustainability are foreseeable advantages of listing, which can help SMEs attract foreign investments. Nevertheless, the extent to which SMEs can benefit from the listing is still unanswered and requires further research. Besides, the challenges created by COVID-19 raise multiple questions on the sustainability of small and medium firms. According to an estimation by the International Monetary Fund, the failure rate of SMEs under COVID-19 is nearly nine percentage points.¹ In addition, the inclusion of the impact of an event like COVID-19 on capital structure choices can be an area of research.

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Note

1. <https://www.imf.org/en/Publications/WP/Issues/2020/09/25/COVID-19-and-SME-Failures-49753>

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